SHAH MOHAMMAD MAHDI HASAN

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Research Interests

Deep learning and computer vision for medical imaging and healthcare, with a focus on automated disease detection, diagnosis, and medical image analysis. Particular interests include explainable AI (XAI), multimodal learning, vision transformers, self-supervised learning, and the development of transparent and clinically useful AI systems for early disease diagnosis.

Education

M.Sc. (Engineering) in Information & Communication Technology *2023*
Islamic University, Kushtia, Bangladesh CGPA: 3.48/4.00 | Rank: 4/29

Academic Session: 2020–2021

- Project: *Comparative Study of Deep Learning Approaches for Skin Cancer Classification and Detection*
- Supervisor: Prof. Dr. Paresh Chandra Barman
- Relevant coursework: Artificial Neural Networks, Computer Vision, Wireless Sensor Networks, Advanced Digital Communications

B.Sc. (Honours) in Information & Communication Technology *2022*
Islamic University, Kushtia, Bangladesh CGPA: 3.25/4.00

Academic Session: 2016–2017

- Final Year Project: *Brain Tumor Detection Using Convolutional Neural Networks (CNN)*
- Supervisor: Prof. Dr. Paresh Chandra Barman
- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Artificial Intelligence & Expert Systems, Digital Image Processing, Digital Signal Processing, Computer Architecture, Computer Networking

Research Experience

Deep Learning for Skin Cancer Classification and Detection *M.Sc. Project*

- Conducted a comparative study of deep learning architectures for multi-class skin cancer classification using dermoscopic images.
- Investigated VGG16, VGG19, ResNet50, ResNet101, ResNet152, DenseNet169, DenseNet201, MobileNetV2, Xception, and InceptionV3.
- Developed preprocessing, normalization, data augmentation, training, and evaluation pipelines using Python and deep learning frameworks.
- Evaluated models using accuracy, precision, recall, F1-score, and Cohen's kappa.
- Best-performing model: DenseNet169, achieving approximately 87.42% accuracy, 87.30% F1-score, and 85.85% Cohen's kappa.
- Prepared a research manuscript based on the project for potential IEEE conference submission.

Brain Tumor Detection Using Convolutional Neural Networks *B.Sc. Final Year Project*

- Developed a convolutional neural network pipeline for automated brain tumor detection from magnetic resonance imaging (MRI) data.
- Investigated image preprocessing, normalization, augmentation, model training, and hyperparameter tuning.
- Evaluated the model using classification performance metrics including accuracy and sensitivity.
- Developed the project as an application of deep learning and computer vision to medical image analysis.

Manuscripts in Preparation

- *Exploring Deep Learning Techniques for Early Skin Cancer Identification* — based on M.Sc. project; manuscript prepared for potential IEEE conference submission.
- *Automated Brain Tumor Detection Using Convolutional Neural Networks* — based on B.Sc. final year project; manuscript prepared for potential conference submission.

Technical Skills

Programming	Python, C++, C, R, MATLAB, SQL, Java
Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, Spark ML
Computer Vision	OpenCV, image preprocessing, image augmentation, CNNs, medical image analysis
Data Science	Pandas, NumPy, Matplotlib, statistical analysis, data preprocessing
Tools	Git, Jupyter Notebook, Google Colab, VS Code, PyCharm, LaTeX
Research	Deep Learning, Computer Vision, Medical Imaging, Explainable AI, Multimodal Learning

Awards & Achievements

- **Academic Excellence Award**, M.Sc. (Engineering) in ICT, Islamic University — ranked 4th among 29 students.
- **Academic Scholarship**, B.Sc. (Honours) in ICT, Islamic University.

Leadership & Activities

- **Program Coordinator**, Career Club, Islamic University.
- **Executive Member**, IT Society, Islamic University.
- **Member**, SAICT Programming Club, Islamic University.
- **Member**, ICT Innovation Lab, Islamic University.

Standardized Test Scores

TOEFL iBT	115/120	(R: 29, L: 29, S: 28, W: 29)
GRE General Test	338/340	(Q: 170, V: 168, AW: 5.00)

Additional Information

Research Areas	Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Medical Image Analysis, Explainable AI, Multimodal Learning
Future Interests	Vision Transformers, Large Language Models, Medical NLP, Self-Supervised Learning, Graph Neural Networks, Federated Learning, AutoML
Target Degree	Ph.D. in Computer Science / Electrical & Computer Engineering / related fields

References

Prof. Dr. Paresh Chandra Barman
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